

CPSC 483 Project Description

For this course project, you are required to work with a real-world dataset and apply machine learning methods to analyze the data and address a meaningful problem. The project is intended to provide hands-on experience with the complete machine learning workflow, including data understanding, preprocessing, model development, evaluation, and interpretation of results.

The project will be completed in multiple phases, the details of which are outlined in the following section. A comprehensive grading rubric is provided at the end of this document to guide your work and ensure transparency in evaluation.

Prior to initiating the project, you and your team must select a **real-world dataset** and a corresponding problem or research question that you would like to investigate. All subsequent phases of the project will be developed based on this selected dataset and problem.

Phase 1: Problem Analysis and Dataset Selection (Teamwork)

The initial stage of the machine learning project involves identifying a meaningful real-world problem and selecting an appropriate dataset for analysis. In a typical machine learning application, the problem may arise from a practical need in areas such as healthcare, finance, transportation, business, science, or social applications. For the purpose of this project, you and your team are expected to conduct independent research and identify a problem that can be reasonably addressed using machine learning techniques.

Your team must select a real-world dataset that is relevant to the problem you would like to investigate. The dataset should contain sufficient information and complexity to support meaningful analysis and machine learning model development. The scope of the dataset must remain balanced: *it should not be overly simplistic, nor should it be unnecessarily large or complicated*. Keep in mind that larger and more complex datasets generally require additional time and computational resources for data preprocessing, model training, testing, and evaluation.

You are encouraged to carefully examine the dataset before making your final selection. Consider the number of observations, available features, data types, missing values, and the type of machine learning problem that can be formulated from the data. Your selected dataset should allow your team to demonstrate meaningful application of the concepts and techniques discussed throughout this course.

Upon completion of Phase 1, the first section of your **Project Final Report** will be prepared. This section must include:

1. **Problem Introduction** – Provide an overview of the real-world problem your team intends to investigate. Explain the motivation for selecting this problem and why it is meaningful or useful to study.
2. **Dataset Description** – Provide a clear description of the selected dataset, including the type of data it contains, the approximate number of observations and features, and other important characteristics of the dataset.
3. **Dataset Source** – Clearly identify where the dataset was obtained and provide a direct link to the original dataset or data repository. Examples may include Kaggle, UCI Machine Learning Repository, government open-data portals, or other publicly available and reliable sources. Some available sources of data:
 - a. Kaggle (<https://www.kaggle.com/datasets>)
 - b. UCI Machine Learning Repository (<https://archive.ics.uci.edu/datasets/>)
 - c. Data.gov (<https://data.gov/>)
 - d. Dataset - California Open Data (<https://data.ca.gov/dataset>)
4. **Proposed Machine Learning Task** – Briefly describe how you expect to use the dataset.

This section will establish the problem context and dataset foundation for all subsequent phases of the project.

Phase 2: Data Preprocessing (Teamwork)

The second phase of the project focuses on **data preprocessing and preparation**. Real-world datasets are rarely ready to be used directly for machine learning. They may contain missing values, inconsistent formats, duplicate records, irrelevant features, categorical variables, outliers, or other issues that can affect the quality and reliability of the analysis.

In this phase, you and your team are expected to carefully examine the selected dataset and perform the preprocessing steps necessary to prepare it for subsequent machine learning tasks. The specific preprocessing techniques will depend on the characteristics of your dataset and the problem being investigated. You should be able to explain what preprocessing steps were performed and why they were necessary.

Possible preprocessing tasks may include:

- Handling missing values
- Removing duplicate or irrelevant records

- Correcting inconsistent data formats
- Encoding categorical variables
- Scaling or normalizing numerical features
- Identifying and addressing outliers
- Selecting or transforming features

Note that not every technique will be appropriate for every dataset; therefore, your team should apply only the methods that are relevant and justify your decisions.

You should also begin separating the data appropriately for model development. For supervised learning problems, this will typically include dividing the dataset into training and testing sets. Depending on the machine learning methods used later in the project, additional validation data or cross-validation techniques may also be considered.

Upon completion of Phase 2, the second section of your **Project Final Report** should include:

1. **Data Cleaning** – Explain any steps taken to address missing values, duplicates, inconsistent records, incorrect values, or other data quality problems.
2. **Data Transformation** – Describe any transformations applied to the data, such as categorical encoding, normalization, standardization, or feature transformation.
3. **Feature Preparation** – Identify the features that will be used for the machine learning analysis and explain any features that were removed, created, or modified.
4. **Training and Testing Preparation** – Explain how the dataset was divided or otherwise prepared for model training and evaluation. Include the training/testing proportions when applicable.

All preprocessing decisions should be clearly documented and supported by appropriate reasoning. This phase will ensure that the dataset is properly prepared for the machine learning methods applied in the subsequent phases of the project.

Phase 3: Model Selection, Development, and Evaluation

Only this phase of the project will be graded individually.

In this phase, each group will be individually responsible for selecting, implementing, and analyzing one machine learning model/algorithm per member that is appropriate for the problem identified in the previous phases.

The selected models should be consistent with the type of machine learning problem being investigated. For a supervised learning problem, you can select algorithms that fall within the

supervised learning category, such as linear or logistic regression, decision trees, random forests, support vector machines (SVM), k-nearest neighbors (KNN), neural networks, or other appropriate supervised learning methods. For an unsupervised learning problem, you can select algorithms that fall within the unsupervised learning category, such as clustering methods, dimensionality reduction techniques, Principal Component Analysis (PCA), or other appropriate unsupervised learning methods.

Each group member should select a different model or algorithm whenever possible so that the group can explore and compare multiple approaches to the same problem. You are also encouraged to explore machine learning algorithms or software libraries that are not directly covered in class, provided that they can clearly explain the selected method and justify why it is appropriate for the project.

Each student is responsible for developing and analyzing their selected model. The analysis should include:

- **Model Selection and Description** – Identify the selected machine learning model or algorithm and provide a brief explanation of how it works. Explain why the model is appropriate for the selected dataset and problem.
- **Model Implementation** – Describe how the model was implemented, including the major software libraries, important parameters, and any relevant model settings.
- **Model Training** – Explain how the model was trained using the prepared dataset. If applicable, describe any parameter tuning, cross-validation, or other techniques used during model development.
- **Model Evaluation** – Evaluate the performance of the model using at least three appropriate evaluation measurements or methods. The selected measurements should be suitable for the type of machine learning problem and model being used.
 - For example, evaluation measurements for supervised classification problems may include accuracy, precision, recall, F1-score, ROC curve/AUC, confusion matrix, or other appropriate measurements. Regression problems may use measurements such as MAE, MSE, RMSE, or R^2 .
 - For unsupervised learning problems, students should select appropriate evaluation approaches based on the selected algorithm, such as clustering quality measurements, reconstruction error, explained variance, or other suitable methods.
- **Results and Analysis** – Present and interpret the results produced by the model. Appropriate tables, figures, plots, or visualizations should be included when necessary. Do NOT simply report numerical results; explain what the results indicate about the performance, strengths, and limitations of the selected model.

Individual Grading

Each group member is responsible for the completeness and quality of the model assigned to them.

If the requirements are not satisfied for a particular model—for example, if the model is not properly implemented, fewer than three appropriate evaluation measurements are provided, or the analysis is incomplete—only the student responsible for that model will receive the corresponding point deduction. Other group members will not lose points for deficiencies associated with another student's assigned model.

Each student's contribution should therefore be clearly identified **in the beginning of the Project Report**, including each member's name and the model for which they are responsible.

Final Project Deliverables

The final project deliverables consist of three primary components: **Project Design Report, Implementation, and Presentation**. Together, these constitute 100% of the final project grade (30% of the overall course grade for undergraduate students and 25% for graduate students).

Project Design Report (60% of Project Grade)

The Project Design Report must follow the template provided for formatting and organization. The report will be evaluated on clarity, completeness, and adherence to the design principles covered in class. It must include the following sections:

Introduction

- Provide a brief description of the selected real-world problem and project topic.
- Explain the motivation for selecting the problem and why it is meaningful to investigate.
- Provide the **source and direct link** to the original dataset.
- Clearly identify the proposed machine learning task, such as classification, regression, clustering, dimensionality reduction, or another appropriate task.

Data Preprocessing

- Describe the initial condition and structure of the dataset.
- Identify any data quality issues, including missing values, duplicate records, inconsistent values, outliers, or irrelevant features.

- Explain all preprocessing steps performed on the dataset.
- Describe any categorical encoding, normalization, standardization, feature transformation, feature selection, or other preprocessing techniques used.
- Explain how the dataset was divided into training and testing sets, when applicable.

Model

- Each group member must be responsible for selecting, implementing, and analyzing one machine learning model or algorithm.
- The selected model must be appropriate for the type of machine learning problem being investigated.

Conclusion

Summarize the entire project, highlighting the contributions, strengths, and potential future path.

Implementation (30% of Project Grade)

The Implementation portion of the project should contain the complete source code required to reproduce the machine learning analysis presented in the Project Report.

Source Code

- Submit **all source code** used for data preprocessing, model training, testing, evaluation, visualization, and other components of the project.
- The source code should be clearly organized, readable, and appropriately commented.
- The submitted code should be capable of reproducing the major results presented in the Project Report.
- You may also include a README file to provide instructions for running the code, if helpful.
 - If the code is hosted in a public repository or online platform, such as GitHub or Google Colab, you may provide a direct link instead of submitting the files separately.
 - If your code is hosted on an external platform, please make sure that the link remains accessible and that the required permissions are enabled at the time of grading.

Software and Libraries

Please make sure to clearly identify the programming language, software packages, machine learning libraries, frameworks, and other tools used in the project.

- You and your team may use openly available software libraries such as Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, or other appropriate tools.

You are permitted to reuse or modify code from open-source projects, textbook examples, publicly available tutorials, and software libraries. HOWEVER, any externally obtained code or resources **MUST** be properly identified and cited.

- The final project must demonstrate a meaningful contribution from the team and CANNOT consist solely of reproducing an existing project.
- Typically, the project contribution should take one of the following forms:
 - Apply established machine learning algorithms to a new or under-studied dataset or problem. In this case, the contribution comes from the application, preprocessing, analysis, model development, and interpretation of the results.
 - Develop, modify, or extend an approach for an existing dataset or problem. In this case, the contribution may involve modifying an existing method, combining techniques, introducing additional preprocessing or modeling procedures, or developing a new approach.

The team should clearly distinguish between components obtained from external sources and components developed or modified as part of the project in the report.

Submission Requirements

- Compress all files into a single **ZIP** or **TAR** archive.
 - 1 Final report (in PDF)
 - 1 folder include all source code
 - OR link to a public repository with code
- Only the **final uploaded file** will be graded, so ensure that it is complete and correct.
- Test all code thoroughly prior to submission to confirm correctness.

Grading Rubric

Category	Item/Feature	Percentage
Report (60%)	Problem Analysis (Novelty of the problem)	10
	Dataset Description + Dataset Source	10
	Data Preprocessing	20
	Model Development and Evaluation	20
Implementation (30%)	Source Code Completeness and Organization	10
	Reproducibility and Accessibility	10
	Technical Implementation and Contribution	10
Presentation (10%)	Content and Clarity	5
	Organization and Time Management	5